RAJALAKSHMI ENGINEERING COLLEGE

Lavanya V 250701386-2025-CSE

L2

Started on	Monday, 31 August 2026, 9:42 PM
State	Finished
Completed on	Monday, 31 August 2026, 9:45 PM
Time taken	3 mins 29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int n,x;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    scanf("%d",&x);
    for(int i=0;i<n;i++)
    {
        int target=x-arr[i];
        int left=i+1,right=n-1;
        while(left<=right)
        {
            int mid=(left+right)/2;
            if(arr[mid]==target)
            {
                printf("%d\n",arr[i]);
                printf("%d\n",arr[mid]);
                return 0;
            }
            else if(arr[mid]<target)
                left=mid+1;
            else
                right=mid-1;
        }
    }
    printf("No\n");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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